Abhinita Sanabada

ML and AI Engineer

Sunnyvale, CA | (669) 241-8492 | abhinita.sanabada@gmail.com | Portfolio: sanabada.github.io | LinkedIn | GitHub

Seeking full-time ML engineering and AI engineering roles. Available to start now.

SUMMARY

ML and AI engineer who builds LLM agents, retrieval systems, guardrailed AI workflows, and on-device inference, with a software engineering background at Wells Fargo, JPMorgan Chase, and SAP Labs. M.S. Applied Data Intelligence candidate at San José State; former AI engineering intern at a healthcare startup. Focused on AI systems that are evaluated, observable, and safe to run in production.

SKILLS

- **LLM / GenAI:** LangChain, LangGraph, RAG, embeddings and semantic search, MCP, tool calling, multi-agent workflows, NeMo Guardrails, human-in-the-loop design
- **Model serving and training:** vLLM, LoRA fine-tuning, PyTorch, scikit-learn, Spark ML, CNNs, BiLSTMs, attention, self-consistency sampling
- **Evaluation:** calibration and held-out test splits, Macro-F1, class-imbalance handling (focal loss, augmentation), error analysis, benchmarking
- **Data and infrastructure:** Python, SQL, Spark/PySpark, Kafka, Airflow, Snowflake, FastAPI, Docker, Kubernetes, AWS, Azure, Redis, MongoDB, PostgreSQL

AI AND ML PROJECTS

EdgeSupport: on-device IT triage agent **demo video** **code** *HP Edge AI Hackathon, SJSU, 2026*

- Built an IT help-desk agent that runs Qwen2.5-7B locally with vLLM on an HP ZGX Nano (NVIDIA GB10), so private device logs never leave the site.
- Designed self-consistency routing: 3 to 5 samples per ticket, kept on site only when answers agree and cite telemetry evidence; otherwise escalated with a reason code.
- Built the privacy-redaction and telemetry pipeline and led team integration (11 of 16 commits); reached 95% accuracy vs. 75% for a rules-only baseline on a held-out set, with zero PII leaked.

Agentic Recruiting Workflow Simulation **code** *Kafka, FastAPI, LangChain, Redis, AWS*

- Built agents for resume parsing, job matching, and outreach drafting with human approval, on Kafka pipelines with idempotent consumers; benchmarked on 10K records.

Speech Emotion Recognition with Dialogue Context **code** *PyTorch, CNN, BiLSTM, Attention*

- CNN, BiLSTM, and attention over Log-Mel and MFCC features with focal loss for class imbalance; 52.4% Macro-F1 and 55.9% accuracy on IEMOCAP, beating MLP and transformer baselines.

Healthcare Affordability Semantic Search **code** *PySpark, Spark ML, HDFS, MiniLM*

- Cleaned 13.6M+ rows of CMS pricing data into ~47K records with Spark; added MiniLM semantic search and Spark regression price estimates.

EXPERIENCE

Healthcare Startup (Confidential) | *AI Engineer Intern* Jun 2026 – Aug 2026

- Built multi-agent LangChain/LangGraph workflows for patient-data and clinical-document processing.
- Added NVIDIA NeMo Guardrails safety checks and structured outputs; built ingestion and feature-extraction for downstream ML.

Wells Fargo | *Senior Software Engineer* Nov 2021 – Jan 2024

- Prototyped an LLM and RAG ops copilot over runbooks, Splunk, and ServiceNow to help triage production breaks, with human approval.
- Owned Java/Spring services and SQL processing for futures trade, margin, and risk; tuned stored procedures for ~30% faster runtimes.

SAP Labs | *Software Development Engineer 2* Apr 2019 – Oct 2021

- Delivered an automotive orchestration solution on SAP HANA Cloud; owned production reliability and a Kubernetes POC.

JPMorgan Chase & Co. | *Software Engineer* Jan 2017 – Apr 2019

- Built Spring Boot services and Airflow/Spark ETL with data-quality checks for asset and wealth management.

Capgemini | *Software Engineer* Apr 2014 – Aug 2016

- Built banking reports in Core Java, Oracle SQL, and Power BI/Tableau.

EDUCATION

M.S. Applied Data Intelligence – San José State University Expected May 2027

B.Tech. Electronics and Communication Engineering – Biju Patnaik University of Technology 2008 – 2012

Certifications: Machine Learning Engineering (Andrew Ng), AWS Fundamentals, Microsoft Azure Fundamentals